Elton Chang

(510) 513-2561 | trenan2004@gmail.com | https://eltonchang1.github.io | https://github.com/EltonChang1

EDUCATION

Carnegie Mellon University May 2026
Master of Science in Data Analytics for Science, GPA: 3.97/4.0 Pittsburgh, PA

University of California, Santa Cruz Jun 2025
Bachelor of Science in Computer Science, GPA: 3.7/4.0 Santa Cruz, CA

EXPERIENCE

Co-Founder 2026 - Present
AMOS

- Built the Rust/Axum control layer for an AI data analyst that verifies SQL against a parsed safe subset, enforces role-based permissions, runs statistics outside the LLM, and links each claim to evidence for review and replay.
- Designed a versioned REST API (OpenAPI, bearer auth, idempotent SQLite transactions) and packaged it as a non-root Docker image with hardened Docker Compose; GitHub Actions gates tests and benchmarks.

Founder / Technical Lead Aug 2025 - Present
Ashe System Pittsburgh, PA / San Francisco, CA

- Developed a full-stack aircraft-on-ground (AOG) risk platform: a Python/pandas scoring pipeline, FastAPI REST APIs, and a Next.js/React/TypeScript dashboard that turns operator CSV exports into explainable risk scores.
- Improved validated Precision@20 for XGBoost AOG risk ranking from 35% to 62%; tuned hyperparameters and early-warning thresholds with Optuna, increasing median warning lead time from 6.5 to 10.2 days.

Infrastructure Engineer Intern Jul 2023 - Sep 2023
Wintec Industries Inc. Newark, CA

- Reduced configuration drift across 15+ non-production hosts by provisioning test environments with Terraform.
- Cut anomaly detection time by 40% with telemetry, alerting, and dashboards for backup and replication pipelines.

Software Engineer Intern Jul 2022 - Sep 2022
Wintec Industries Inc. Newark, CA

- Built a cloud-hosted internal file management system with Java, SQL, and scheduled Python jobs that automate file ingestion, metadata indexing, and data lifecycle tracking.
- Improved onboarding and adoption by 35% by documenting its architecture and workflows for developers and users.

PROJECTS

MarketPulse AI

- Launched a stock research app (React, Express, MongoDB) on Vercel and Google Cloud Run with JWT auth and an OpenAI tool-calling assistant whose tools also run as a Model Context Protocol (MCP) server.

Enterprise AI Assistant

- Built a multi-tenant retrieval-augmented generation (RAG) assistant (React, Node.js, SQLite) with OpenAI embeddings, cosine-similarity search, function-calling tools, WebSocket streaming, and role-based access control.

Zoe

- Created a social ranking app with an Expo/React Native (TypeScript) client and a Hono/Prisma API on PostgreSQL with argon2id sessions, Google/Apple sign-in, S3 presigned uploads, and full-text search.

Torflix

- Implemented an asyncio BitTorrent engine (peer wire protocol, HTTP/UDP trackers, piece selection) behind a FastAPI/WebSocket daemon; a React/TypeScript PWA plays video via HTTP range requests while downloading.

OPEN SOURCE CONTRIBUTIONS

- Merged 10 pull requests into open-source projects, including Apache DataFusion (Rust SQL engine), SQLFluff (SparkSQL parser), Career Ops (JavaScript), and HoloViz Panel (three UI-state fixes).
- Opened inference-infrastructure pull requests to vLLM TPU Inference (KV-cache connector), Dynamo (VRAM-aware pytest workers), ONNX Runtime (CUDA CMake), and vLLM Production Stack (Kubernetes probe).

TECHNICAL SKILLS

Languages & Web: Python, TypeScript, JavaScript, Rust, Java, Go, C/C++, SQL, React, Next.js, Node.js, FastAPI
AI, Data & Cloud: LLM APIs, RAG, MCP, PyTorch, PostgreSQL, MongoDB, Docker, Kubernetes, Terraform, GCP